

E-Kart

Electric GoKart and Outreach Project Definition



**ELECTRICAL AND
COMPUTER ENGINEERING**
COLORADO STATE UNIVERSITY

Electrical Engineering:

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Computer Engineering:

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Mechanical Engineering:

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Summary

The EGoKart project is a multi-year senior design effort focused on showcasing interdisciplinary engineering collaboration through the design, refinement, and demonstration of an electric go-kart. Electrical Engineering students address wiring, power electronics, and motor control systems. Computer Engineering students focus on embedded systems, Raspberry Pi dashboards, CAN bus communication, and perception technologies. Mechanical Engineering students contribute structural design, steering and drive train improvements, and custom mounts.

The 2025–26 team will continue building on prior progress by finalizing a robust dashboard (hardware and software), improving the reliability of the electrical and drive systems, refining steering, and exploring integration of additional sensors toward autonomous perception. The project highlights teamwork across engineering domains while producing a demonstrable, outreach-ready electric kart that engages younger students and the community in STEM fields.

Why is This Project Important?

This project demonstrates the real-world interdisciplinary teamwork required in engineering practice. By combining mechanical, electrical, and software components into a working electric go-kart, the project provides a tangible and exciting platform for outreach, student learning, and experimentation with different technologies.

Revision History

Date	Comments	Version	Approved by
9/24/25	Initial project plan	1.0	TBD

Problem Statement

The EGoKart project started as a showcase of interdisciplinary engineering, designed to demonstrate how electrical, computer, and mechanical systems integrate into a functional go-kart. Over the years the kart has been refined in electrical systems, motor controls, and software dashboards, but challenges remain.

For 2025-26, the continuation team inherits a kart that is somewhat operational but requires significant mechanical improvements, safety improvements, robustness, and long-term maintainability. In addition, the dashboard is being built from the ground up to make it easier for future teams to work on it and add features. The dashboard electronics are not vehicle-grade and the team are exploring the possibility of redesigning using a PCB-based or ruggedized prototyping methods. Mechanically, steering function, real axle wear, and weak motor mounts must be addressed. Long-term, the kart aims to serve as a testbed for autonomous perception research in collaboration with Prof. Pasricha's Multi-Modal Perception group.

Electrical Engineering Objectives

The primary goals for the Electrical Engineering disciplines will be focused around re-wiring the circuitry of the EGoKart. Accomplishing this will provide a more stable circuit with less power loss, safer and reliable driving performance with secured wiring. Working with the Mechanical Engineering member, the Electrical Engineering members must be able to wire a new circuit around the new infrastructure of the battery storage and motor controls. These objectives will be achieved with more robust wiring, cable management, and efficient circuit design. Additionally, an emergency stop mechanism is an objective to ensure that the driver can shut off power from the battery.

The current state of dashboard circuitry is made with breadboard and simple wiring. The Electrical Engineering team will be developing a PCB-based circuit design for better signal, power management, and system performance.

Computer Engineering Objectives

The Computer Engineering objectives for the EGoKart project center on building a reliable, maintainable, and extensible software systems that interface seamlessly with the kart's electrical and mechanical subsystems. A primary focus will be the development of a robust dashboard application using Qt C++ that replaces the current QML prototype. This dashboard will interface with the Raspberry Pi and CAN bus to display real-time vehicle metrics such as speed, RPM, battery voltage, and temperature. The software must be designed with modularity in mind, allowing future teams to add new features or integrate additional sensors without rewriting the entire codebase.

In addition to dashboard development, Computer Engineering objectives include configuring and testing the MCP2515 CAN controller, establishing stable communication between the Raspberry Pi and motor controllers, and ensuring accurate data parsing from CAN frames. CE students will also collaborate with Electrical Engineering teammates to signal integrity and with Mechanical Engineering teammates to ensure sensor data reflects actual vehicle performance. Longer-term objectives involve preparing the platform for future integration of autonomous perception technologies, including potential support for cameras and machine learning accelerators (e.g., Coral TPU). By meeting these objectives, the CE team will deliver reliable embedded software that ties together the kart's subsystems into a cohesive and demonstrable platform.h

Mechanical Engineering Objectives

The objectives for the mechanical engineers for the EGoKart will be to fix a number of mechanical faults in the current system. An assessment of why the breaks aren't working will need to be done before repairs can be made. Additionally, the rear axles slide off of the EGoKart while turning. A mechanism to prevent this will need to be implemented. A metal box to protect the batteries and contain any possible fire will also need to be made to improve the EGoKarts safety. If there is time the steering mechanism needs to either be reinforced or replaced. Additionally, some components such as the speaker assembly have been simply ziptied to the EGoKart; these will need to be replaced with a more robust and permanent solution.

Electrical Engineering Research

- Kill switch
- Cable management
- PCB Design

Computer Engineering Research

- Qt Creator
- Possibly porting old UI to C++ version
- CAN controller

Mechanical Engineering Research

- Metal battery box
- Drive train

Preliminary Requirements

- **Fix rear Axles so they no longer fall off when turning**
- **Get breaks working**
- **Perform cable management**
- **Transition dash UI from python & QML to C++ & QML**
- **Integrate metrics into dash. Battery charge, electronics temperature, current**
- **Design new stable circuits within dashboard**
- **Design and make a metal box for the batteries**
- **Integrate a kill switch for safety**

Budget Justification

The current known budget for the project is \$200 per senior design student for an expected total of \$1400. There is a possibility for some funds from the previous year to roll over, however; we have not received any verification for this at the present. The bulk of the budget for this semester is expected to be for the mechanical components of the EGoKart. Not everything for purchasing has been decided yet because further analysis of what parts from the current design need replacing. One purchase that is currently planned however is a more adequate ratchet set, as the current tools lack a deep enough socket to reach some of the nuts on the EGoKart. This is expected to be between \$100 and \$200 dollars.

Computer Engineering (CE) Purchases

Item	Qty	Est. Unit Cost	Est. Shipping	Est. Total
Raspberry Pi 5 (8GB)	1	\$80	\$15	\$95
7" Touchscreen Display	1	\$90	\$15	\$105
CAN Bus Interface (MCP2515 module or USB-CAN adapter)	2	\$40	\$10	\$90
MicroSD Cards (64GB, Class 10)	2	\$15	\$5	\$40

CE Subtotal: \$330

Electrical Engineering (EE) Purchases

Item	Qty	Est. Unit Cost	Est. Shipping	Est. Total
VEESC Motor Controller (spare/replacement)	1	\$150	\$20	\$170
High-gauge Wiring & Connectors Kit	1	\$80	\$15	\$95
Sensor Modules (Temp, Voltage, Current)	3	\$30	\$10	\$100
Fuses & Relays Assortment	1	\$50	\$10	\$60

EE Subtotal: \$425

Mechanical Engineering (ME) Purchases

Item	Qty	Est. Unit Cost	Est. Shipping	Est. Total
Ratchet & Deep Socket Set	1	\$150	\$20	\$170
Bearings (Rear Axle Replacement)	4	\$30	\$10	\$130
Steel Tubing / Welding Materials (for steering/mounts)	Bulk	\$120	\$40	\$160
3D Printing Filament / Resin (for brackets & housings)	5 rolls	\$25	\$15	\$140
Motor Mount Plates (custom fabricated or pre-cut)	2	\$75	\$20	\$170

ME Subtotal: \$770

Total estimated cost: \$1525

Process Step	Potential Failure Mode	Potential Failure Effect	SEV1	Potential Causes	OCC2	Current Process Controls	DET 3	RPN4	Action Recommended
What is the step?	In what ways can step go wrong?	What is the impact on the customer if the failure mode is not prevented or corrected?	How severe is the effect on the customer?	What causes the step to go wrong (i.e. how could the failure mode occur)?	How frequently is the cause likely to occur?	What are the existing controls that either prevent the failure mode from occurring or detect it should it occur?	How probable is detection of the failure mode or its cause?	Risk priority number calculated as SEV x OCC x DET	What are the actions for reducing the occurrence of the causes or for improving its detection? Provide actions on high RPNs and on severity ratings of 9 or 10
Start the Kart	motor doesn't turn	Kart doesn't move	10	Can bus failure, motor controller shortage, wiring disconnected	4	correctly soldered parts, vsc troubleshooting	1	40	bondic wires together
	Pedal disconnects	Kart doesn't move	10	Wiring shorts, parts disconnect	2	Heat shrink connectors, Bondic	1	20	heat shrink and bondic
Charging	Battery Failure	Fire. Batteries explode.	10	Incorrect charging method. Overcharging/Unregulated charging. Cell failure	3	C240 Smart Charger, but requires 24/7 monitoring	2	60	Battery/ESC/Motor setup has better cable management and dedicated charging mode.
Raspberry pi	can bus doesn't connect	UI will not load	4	Poor physical connection, issue	2	It will not turn on if the can bus is not linked with each other. another	3	24	

				from either motor controller		good sign is the motors aren't linked together			
	Camera doesn't connect	The UI may load but our model does not	6	Poor connections, code is not run properly	5	Raspberry pi are pre programmed, you will know if the camera doesn't turn on	2	60	Make sure the bash script will always load. Backup micro-sd to plug and play
	Machine learning model doesn't stop the vehicle	They crash into a pedestrian	9	Model does not recognize the human or the depth of the human	3	Model is pre trained and trained to detect people more often than not	3	81	Extensive testing on the model and the vehicle before deployment in school demo
Full Speed	Brakes stop working	Will not be able to stop Kart	10	The break pressure isn't set correctly	4	Safety testings and pressure gauges	3	120	Breaks tests periodically. Make sure we can stop on a dime
	Motor controller wrong amount of current	The speedometer will be set incorrectly	6	Motor controllers are reading the wrong currents	2	vesc testing, physical speedometer	6	72	Always test the speedometer with a physical and the virtual. Double check the current limiter

Computer and Electrical Engineering Risk And Reliability Assessment:

#	Risk Event	Probability/100	Impact, weeks	Score, hours	Effect	Risk Mitigation Plan	Person responsible for implementing control
1	Battery Failure	40	12	100	Delay, Cost, Project loss, Injury	Updated Safety standards, marrying batteries, sensors and frequent quality checks.	EE Team
2	VESC burns out	30	2	30	Delay, Cost, and Injury	VESC well cooled	EE Team

3	BMS burns out	30	2	100	Delay, Cost, and Injury	BMS needs to support 44.4 V	Whole Team
4	APD doesn't detect pedestrians	50	2	20	Delay, Cost, and Injury	Improve ML model and integrate LiDAR	AI Team
5	CAN bus failure	10	3	20	Delay, Cost	Anomaly detection	CpE and EE team
Total risk :					Delay, Cost, and Injury	Maintaining code and double checking components, maintaining safety standards	

Mechanical Engineering Risk And Reliability Assessment:

#	Risk Event	Probability/100	Impact, weeks	Score, hours	Effect	Risk Mitigation Plan	Person responsible for implementing control
1	Breaks Fail	30	2	10	Delay, Cost, and Injury	Inspect and test breaks before riding	ME Team
2	Steering Fail	30	2	10	Delay, Cost, and Injury		ME Team
3	Electrical Over Heating	40	3	50	Delay, Cost		ME Team
4							
5							
Total risk :					Delay, Cost, and Injury	Maintaining code and double checking components, maintaining safety standards	

Computer Engineering Timeline			
Task	Date Start	Date End	Team Member
Research Porting Old Dash to Qt C++	9/8	9/17	Hoguer
Research Qt Creator C++ & QML For New Dash	9/8	9/22	Nic D
Create Prototype Dashboard	9/22	10/13	Nic D
Update Website	9/10	9/19	Paul
Implement Core & IO			CpE Team
Domain & State			
UI Shell & Tiles			
Testing & Build			
Performance & Polish	11/8	11/14	CpE Team
Migration & Clean Up	11/15	11/21	CpE Team

Electrical Engineering Timeline			
Task	Date Start	Date End	Team Member
Kill Switch	9/24	10/8	Kyle
Wire Management	10/6	10/20	Nick
Control Panel Circuits			
Schematics			All
Powering Subsystem			

Mechanical Engineering Timeline			
Task	Date Start	Date End	Team Member
CAD Model Metal Box	9/1	9/7	Dillon
Disassemble GoKart Rear Assembly	10/1	10/8	Gray, Evan
Assess Axles and Brake System for Parts and Order	10/8	10/15	Gray, Evan
Metal Battery Box	9/22	10/6	Nick
Analyze Break System for Faults			
Break System			
Repair & Reassemble Rear Axles	10/20	11/3	Gray, Evan
Reassembly			